Digital System Design Report (UEC612)

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1. AIM :

To model, simulate, synthesise, and test a 4-bit Synchronous Binary Up/Down Counter with Synchronous Preset and Ripple Carry Output using IC **SN74LS669**, and implement the design on the Spartan-7 Boolean FPGA Board using Xilinx Vivado .

2. TOOLS USED :

Software	Xilinx Vivado
Language	Verilog HDL (IEEE 1364-2001)
FPGA Board	Spartan-7 Boolean Board (Digilent)
Target Device	XC7S50-CSGA324
Reference IC	SN74LS669 — Texas Instruments

3. THEORY :

The SN74LS669 is a fully synchronous, presettable 4-bit binary up/down counter. All four flip-flops are triggered simultaneously on the rising edge of the clock, which prevents the output spikes seen in ripple (asynchronous) counters. Internal look-ahead carry logic is included for fast cascading without extra gating.

- **Synchronous Counting:** The direction is controlled by U/D. When HIGH the counter counts up (0 to 15); when LOW it counts down (15 to 0). Wrapping occurs automatically.
- **Synchronous Preset (LOAD_bar):** Pulling LOAD_bar LOW causes the 4-bit parallel data on pins A–D to be loaded into the counter on the next rising clock edge.
- **Count Enable (ENP and ENT):** Both must be LOW for counting to proceed. If either is HIGH the counter holds its value. ENT also controls the RCO_bar output.
- **Ripple Carry Output (RCO_bar):** Goes LOW at terminal count (15 for up, 0 for down) when ENT is asserted. This output is used to cascade multiple counters.
- **Clock Divider (FPGA):** A 26-bit register divides the 100 MHz board clock down to approximately 1.5 Hz so the LED outputs change at a visible rate.

4. LOGIC DIAGRAM (RTL SCHEMATIC) :

The schematic below was generated by Xilinx Vivado after elaboration. It shows the adder, subtractor, multiplexers, equality comparators for RCO_bar, and the 4-bit register driven by the divided slow clock.

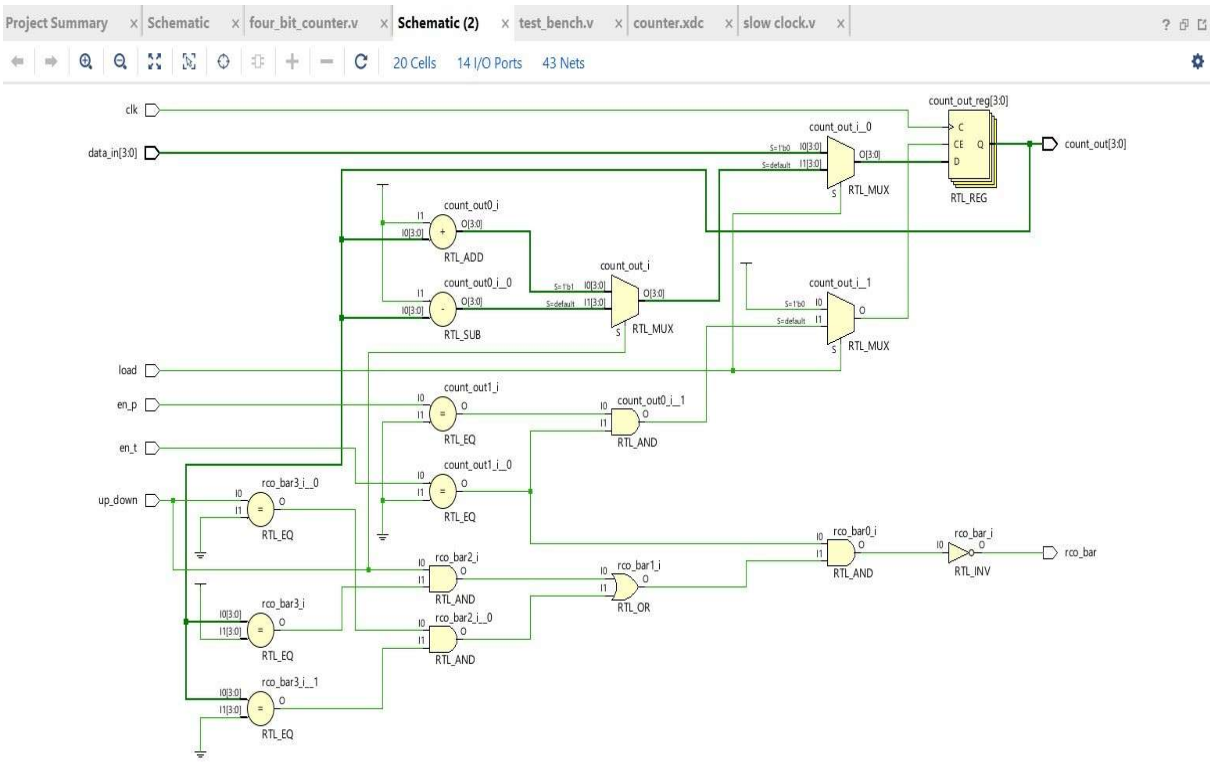

Figure 1: RTL Schematic

5. PIN DESCRIPTION

Pin	Signal	I/O	Description
2	CLK	In	Clock input — all operations trigger on rising edge
9	LOAD_bar	In	Active-LOW synchronous parallel load
1	U/D	In	HIGH = Count Up, LOW = Count Down
7	ENP	In	Active-LOW count enable (parallel)
10	ENT	In	Active-LOW count enable; also gates RC
3–6	A–D	In	4-bit parallel data input (A = LSB, D = MSB)

14	QA	Out	Bit 0 (LSB) of counter output
13	QB	Out	Bit 1 of counter output
12	QC	Out	Bit 2 of counter output
11	QD	Out	Bit 3 (MSB) of counter output
15	RCO_bar	Out	Active-LOW Ripple Carry — LOW at terminal count
8	GND	Pwr	Ground (0 V)
16	VCC	Pwr	Supply (+5 V)

6. FUNCTION TABLE :

H = Logic HIGH, L = Logic LOW, X = Don't Care, ↑ = Rising edge of CLK.

LOAD_bar	ENP	ENT	U/D	CLK	Operation
L	X	X	X	↑	Synchronous Load — data_in loaded into counter
H	H	X	X	↑	Inhibit — counter holds current value
H	X	H	X	↑	Inhibit — counter holds (RCO_bar also forced HIGH)
H	L	L	H	↑	Count Up → 0, 1, 2 ... 14, 15, 0, 1 ...
H	L	L	L	↑	Count Down → 15, 14 ... 1, 0, 15, 14 ...
H	L	L	H	–	RCO_bar = LOW when count = 15 (terminal up count)
H	L	L	L	–	RCO_bar = LOW when count = 0 (terminal down count)

7. TRUTH TABLE :

Count-Up Sequence (U/D = H, LOAD_bar = H, ENP = L, ENT = L) :

Clk Edge	QD	QC	QB	QA	Decimal	RCO_bar
1	0	0	0	0	0	H
2	0	0	0	1	1	H
3	0	0	1	0	2	H
4	0	0	1	1	3	H
5	0	1	0	0	4	H
6	0	1	0	1	5	H
7	0	1	1	0	6	H
8	0	1	1	1	7	H
9	1	0	0	0	8	H
10	1	0	0	1	9	H
11	1	0	1	0	10	H
12	1	0	1	1	11	H
13	1	1	0	0	12	H
14	1	1	0	1	13	H
15	1	1	1	0	14	H
16	1	1	1	1	15	**L**

Count-Down Sequence (U/D = L, LOAD_bar = H, ENP = L, ENT = L)

Clk Edge	QD	QC	QB	QA	Decimal	RCO_bar
1	1	1	1	1	15	H
2	1	1	1	0	14	H
3	1	1	0	1	13	H
4	1	1	0	0	12	H
5	1	0	1	1	11	H
6	1	0	1	0	10	H
7	1	0	0	1	9	H
8	1	0	0	0	8	H
9	0	1	1	1	7	H
10	0	1	1	0	6	H
11	0	1	0	1	5	H
12	0	1	0	0	4	H
13	0	0	1	1	3	H
14	0	0	1	0	2	H
15	0	0	0	1	1	H
16	0	0	0	0	0	**L**

8. VERILOG SOURCE CODE :

Main Module — four_bit_counter.v

```
`timescale 1ns/1ps
module four_bit_counter(
input clk,
input load, // Active-LOW synchronous load
input en_p, // Active-LOW count enable (parallel)
input en_t, // Active-LOW count enable (terminal)
input up_down, // 1 = Count Up, 0 = Count Down
input [3:0] data_in,
output reg [3:0] count_out,
output rco_bar
);

reg [25:0] clk_div;
wire slow_clk;

// Clock Divider: 100 MHz to approx 1.49 Hz
always @(posedge clk)
clk_div <= clk_div + 1;

assign slow_clk = clk_div[25];

// Counter Logic
always @(posedge slow_clk) begin
if (load == 0)
count_out <= data_in;
else if (en_p == 0 && en_t == 0) begin
if (up_down == 1)
count_out <= count_out + 1;
else
count_out <= count_out - 1;
end
end

// Ripple Carry Output
assign rco_bar = ~(
(en_t == 0) && (
```

```
(up_down == 1 && count_out == 4'b1111) ||
(up_down == 0 && count_out == 4'b0000)
)
);

endmodule
```

Testbench — test_bench.v :

```
`timescale 1ns/1ps
```

```
module test_bench;
reg clk, load_bar, en_p, en_t, up_down;
reg [3:0] data_in;
wire [3:0] count_out;
wire rco_bar;

my_counter uut (
.clk(clk), .load_bar(load_bar), .en_p(en_p),
.en_t(en_t), .up_down(up_down), .data_in(data_in),
.count_out(count_out), .rco_bar(rco_bar)
);

always #5 clk = ~clk;

initial begin
clk = 0; load_bar = 1; en_p = 0; en_t = 0;
up_down = 1; data_in = 4'b0000;

// Load decimal 5
#10 data_in = 4'b0101; load_bar = 0;
#10 load_bar = 1;

// Count Up
#40;

// Switch to Count Down
up_down = 0;
#60;
```

```
$finish;
end
endmodule
```

9. XDC CONSTRAINTS FILE :

The XDC file maps design ports to physical FPGA pins and sets the I/O standard.

```
## Clock
set_property PACKAGE_PIN W5 [get_ports clk]
set_property IOSTANDARD LVCMOS33 [get_ports clk]
create_clock -period 10.000 -name sys_clk [get_ports clk]

## Parallel Data Input (Slide Switches SW0-SW3)
set_property PACKAGE_PIN U2 [get_ports {data_in[0]}]
set_property PACKAGE_PIN U1 [get_ports {data_in[1]}]
set_property PACKAGE_PIN T2 [get_ports {data_in[2]}]
set_property PACKAGE_PIN T1 [get_ports {data_in[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {data_in[*]}]

## Push Buttons
set_property PACKAGE_PIN J3 [get_ports load]
set_property PACKAGE_PIN J5 [get_ports en_p]
set_property PACKAGE_PIN H2 [get_ports en_t]
set_property PACKAGE_PIN J1 [get_ports up_down]
set_property IOSTANDARD LVCMOS33 [get_ports load]
set_property IOSTANDARD LVCMOS33 [get_ports en_p]
set_property IOSTANDARD LVCMOS33 [get_ports en_t]
set_property IOSTANDARD LVCMOS33 [get_ports up_down]

## LED Outputs (count_out)
set_property PACKAGE_PIN G3 [get_ports {count_out[0]}]
set_property PACKAGE_PIN G2 [get_ports {count_out[1]}]
set_property PACKAGE_PIN F2 [get_ports {count_out[2]}]
set_property PACKAGE_PIN F1 [get_ports {count_out[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {count_out[*]}]

## RCO Output
```

```
set_property PACKAGE_PIN E1 [get_ports rco_bar]
set_property IOSTANDARD LVCMOS33 [get_ports rco_bar]
```

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10. SIMULATION RESULTS :

Behavioural simulation was performed in Xilinx Vivado using the testbench from . The waveform below confirms all operating modes.

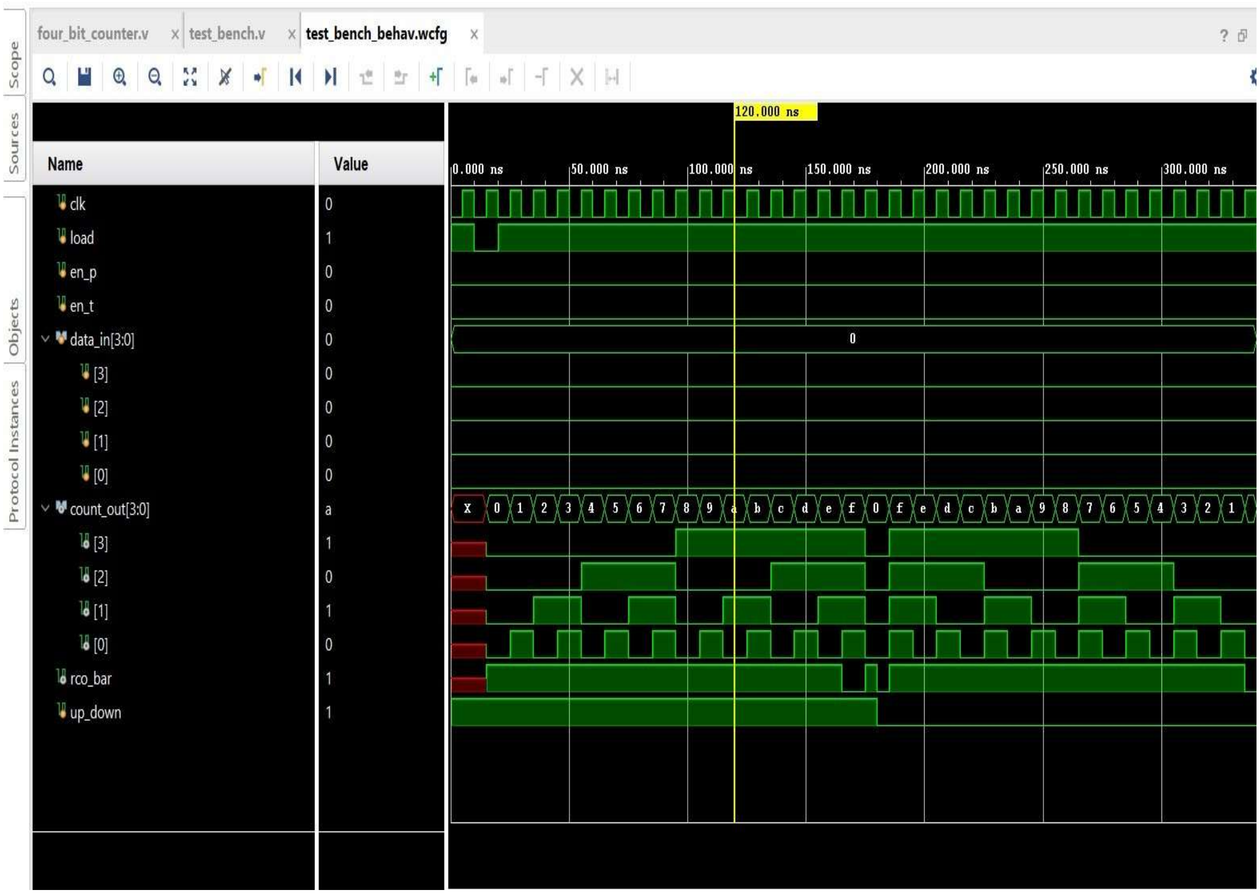

Figure 2: Behavioural Simulation Waveform

Simulation Analysis :

- **0 – 10 ns:** Initialisation. Inputs set to defaults; count_out is undefined (X) before the first clock edge.
- **10 – 20 ns:** Synchronous Load. data_in = 0101 (decimal 5), load_bar pulsed LOW. count_out captures 5 on the next rising edge.
- **20 – 110 ns:** Count-Up phase. With up_down = 1 and both enables LOW, the counter increments each cycle: 5, 6, 7, 8, 9, A, B, C, D, E, F. At count = 15 (0xF), rco_bar pulses LOW indicating terminal count.
- **Wrap-around:** After 15 the counter rolls over to 0 and continues upward, consistent with the 74LS669 binary counter specification.
- **110 – 300 ns:** Count-Down phase. up_down switched LOW. Counter decrements each cycle: F, E, D, ... 1, 0. At count = 0, rco_bar asserts LOW again.
- **RCO behaviour:** Ripple Carry Output correctly asserts at terminal counts in both directions when ENT is enabled, confirming cascading capability.

11. FPGA IMPLEMENTATION

After successful simulation, the design was synthesised, implemented, and the bitstream was programmed onto the Spartan-7 Boolean Board via JTAG using the Vivado Hardware Manager.

FPGA	XC7S50-CSGA324 (Spartan-7)
Clock Pin	W5 — 100 MHz onboard oscillator
Slow Clock	clk_div[25] — approximately 1.49 Hz
LOAD button	Pin J3

ENP button	Pin J5
ENT button	Pin H2
UP/DOWN button	Pin J1
Data Input	Pins U2, U1, T2, T1 (SW0–SW3)
Count Output	Pins G3, G2, F2, F1 (LD0–LD3)
RCO Output	Pin E1 (LD4)
I/O Standard	LVCMOS33 (3.3 V)

12. RESULT AND CONCLUSION :

The 4-bit Synchronous Binary Up/Down Counter based on the SN74LS669 IC was successfully designed in Verilog, simulated in Xilinx Vivado, and implemented on the Spartan-7 FPGA board. The following results were observed:

- The synchronous load correctly captured parallel data (decimal 5) on the next rising clock edge when LOAD was asserted.
- The counter accurately counted up from 0 to 15 and counted down from 15 to 0, wrapping around continuously without any glitches.
- The Ripple Carry Output (RCO) asserted LOW precisely at the terminal counts — decimal 15 for up-counting and decimal 0 for down-counting — confirming correct cascading behaviour as per the 74LS669 datasheet.
- The inhibit function correctly froze the counter when either ENP or ENT was deasserted, with nounintended state changes observed.
- The FPGA implementation with the 26-bit clock divider produced a clearly visible count rate on the LEDs, demonstrating successful hardware deployment.
- All simulation waveforms matched the expected truth table, validating the Verilog model against the Texas Instruments SN74LS669 datasheet.